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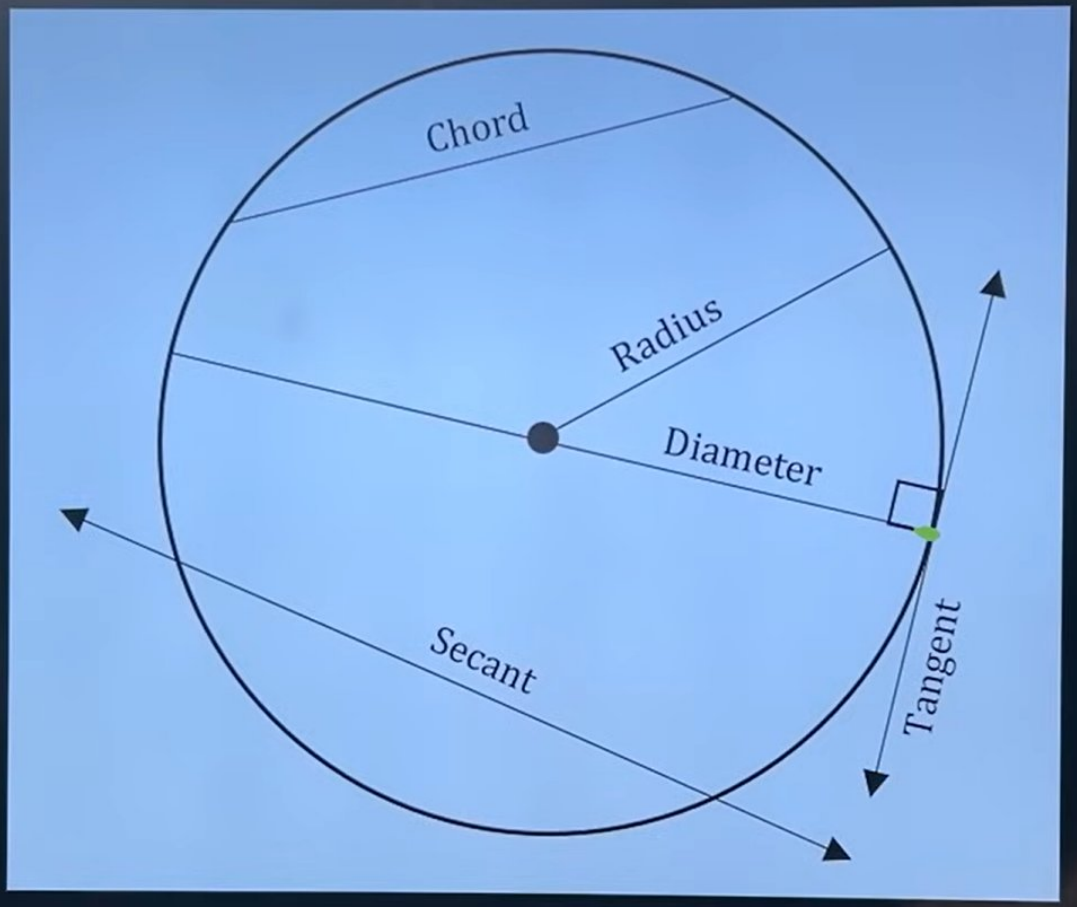
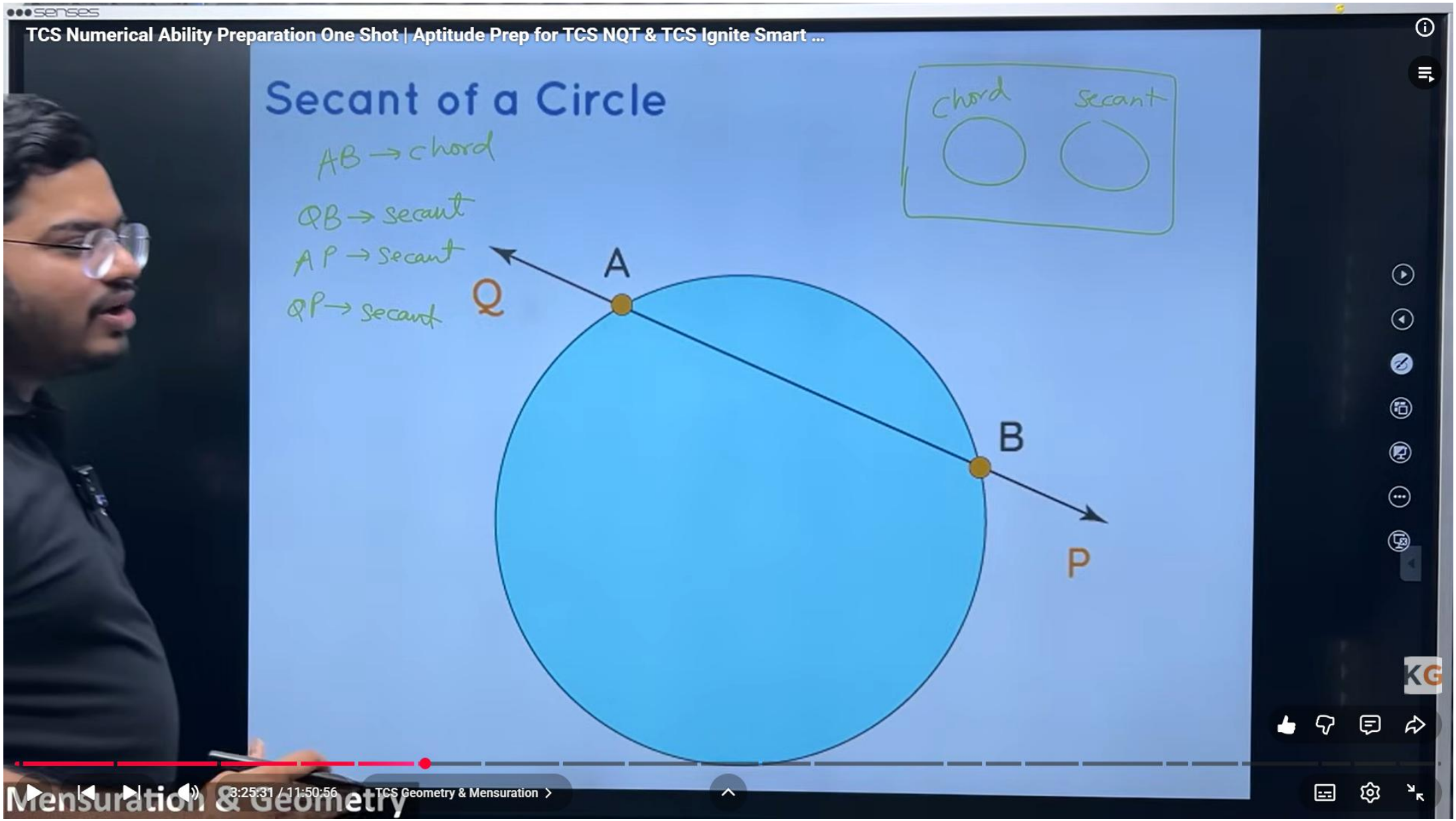


Mensuration & Geometry
3:24:08 / 11:50:56
TCS Geometry & Mensuration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Secant of a Circle
AB → chord
QB → secant
AP → secant
QP → secant
chord
secant
A
B
Q
P
3:25:31 / 11:50:56
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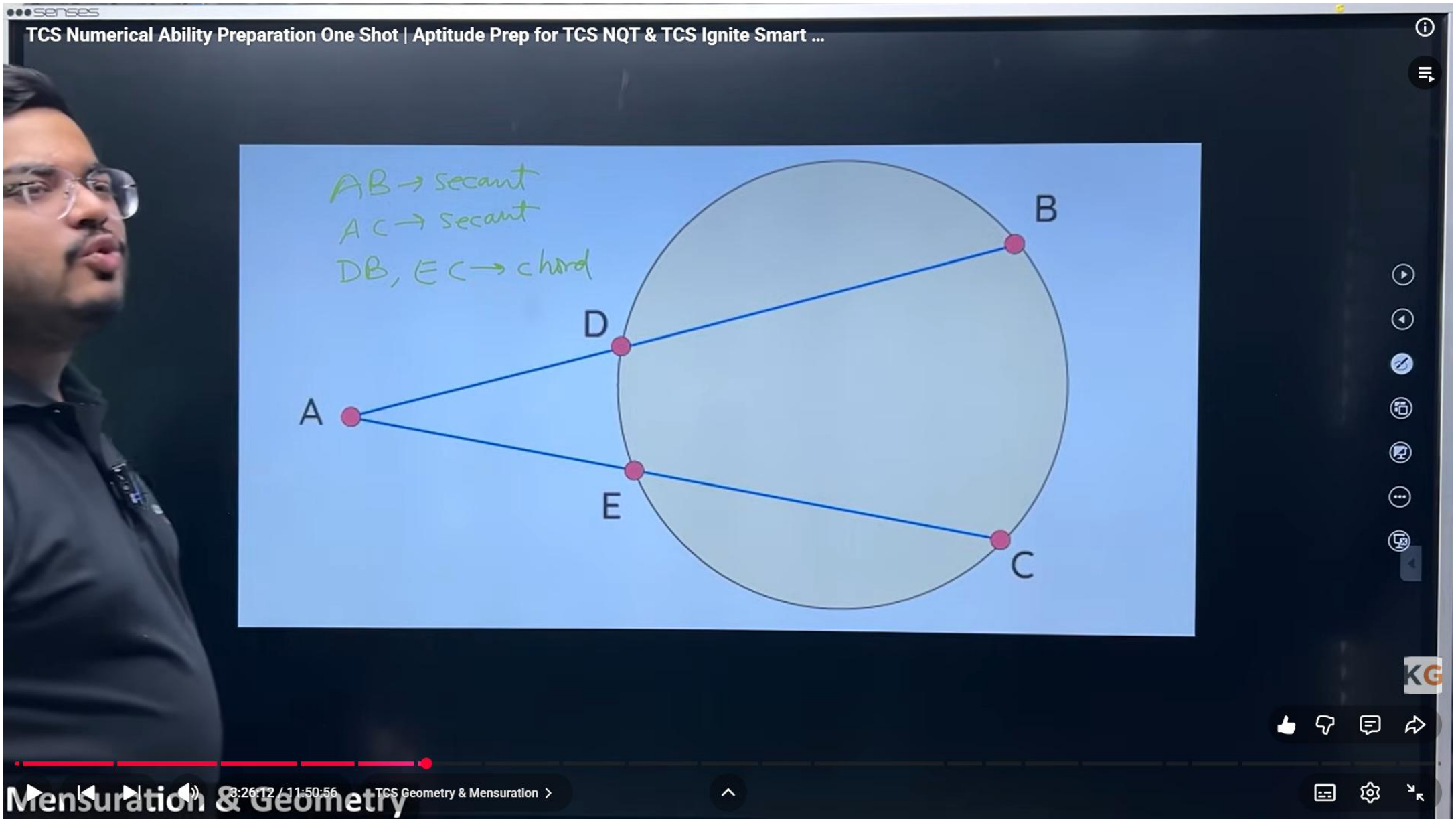
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AB → secant
AC → secant
DB, EC → chord
A
B
C
D
E
3:26:12 / 11:50:56
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Q1. Which of the following statements is incorrect in regard to a circle?

i) Every Chord is Secant ✗
ii) Every Secant is Chord ✗
iii) Secant is technically not a chord, but it contains a chord. ✓
iv) Every secant line defines a unique chord. ✓

a) Only ii
b) Only i ii & iii
c) Only i & ii
d) All are correct

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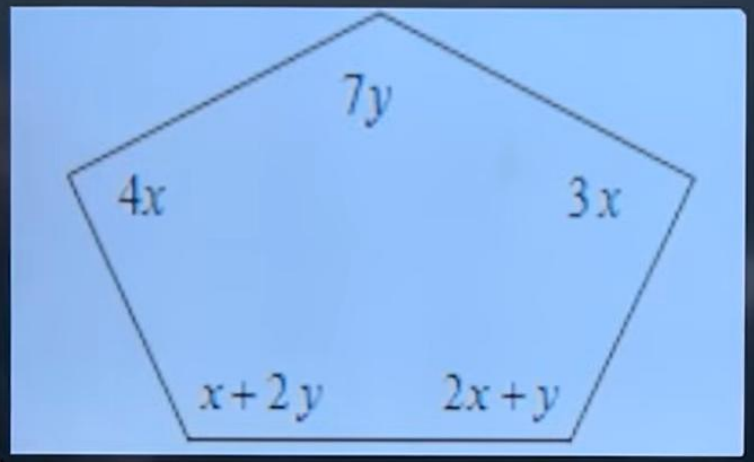
Q2. If the expression shown are the degree measures of the angles if the pentagon, find the value x+y.

a) 36
b) 45
c) 54
d) 63



Interior angle sum = $(n-2)\times 180^\circ$

$\boxed{n=5}$

$= (5-2)\times 180^\circ$

$= 3\times 180^\circ = 540^\circ$

$7y + 3x + (2x+y) + (x+2y) + 4x = 540^\circ$

$10x + 10y = 540$

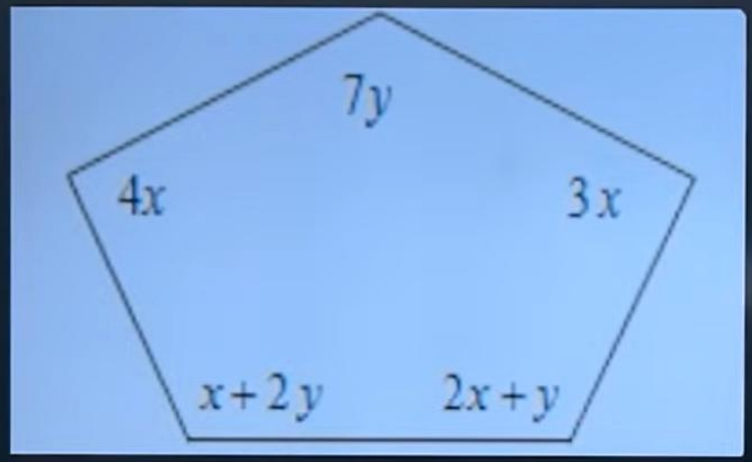
$10[x+y] = 540 \Rightarrow x+y = \frac{540}{10}$

$= 54$

3:30:09 / 11:50:56 TCS Geometry & Mensuration
For Placement Courses by Yash Bhaiya, Checkout KnowledgeGate Website, Link is in Description

Q2. If the expression shown are the degree measures of the angles if the pentagon, find the value x+y.

a) 36
b) 45
c) 54
d) 63



$n = 3$

$(3-2) \times 180^\circ = 180^\circ$

$\frac{180}{3} = 60^\circ$

Interior angles sum $= (n-2) \times 180^\circ$

For regular polygon [all sides & angle are equal]

Individual Angle $= \frac{(n-2) \times 180}{n}$

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Q3. One angle of a regular polygon measures 177° . This polygon has a total on 'n' sides, 'n' is a multiple of which of the following numbers ?

a) 3 ✓ $\frac{120}{3}$

b) 4 ✓ $\frac{120}{4}$

c) 12 ✓ $\frac{120}{12}$

d) All of the above ✓✓

Individual angle $= \frac{(n-2) \times 180}{n}$

$$177 = \frac{(n-2) \times 180}{n}$$

$$177n = (n-2) \times 180$$

$$177n = 180n - 360$$

$$360 = 180n - 177n$$

$$360 = 3n$$

$$n = \frac{360}{3} = 120$$

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Q4. Find the sum of the measures of one interior and one exterior angle of a regular 940-gon.

a) 168°

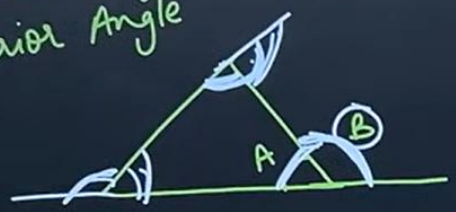
b) 174°

c) 180°

d) 186°

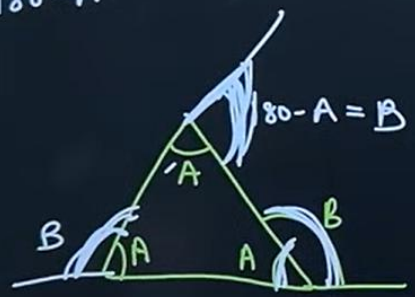
A: Interior Angle

B: Exterior Angle



$A + B = 180$

$B = 180 - A$



$A + B = 180^\circ$

$\boxed{\text{Interior} + \text{Corresponding Exterior} = 180^\circ}$

↳ valid for all polygons [regular, irregular]

$\boxed{\text{Any Interior} + \text{Any Exterior} = 180^\circ}$

↳ valid only for regular polygons

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Q5. What is the measure of the radius of the circle inscribed in a triangle whose sides measure 8, 15 and 17 units?

$b = 8 \quad p = 15 \quad h = 17$

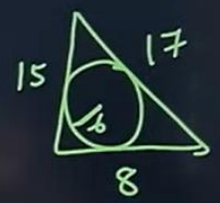
In right angled triangle, $h^2 = b^2 + p^2$

$(17)^2 = (15)^2 + 8^2$

$289 = 225 + 64$

$\boxed{289 = 289}$

$r = \frac{bp}{b+p+h}$



$= \frac{8 \times 15}{8+15+17} = \frac{8 \times 15}{40}$

$= 3$

$r = \frac{b+p-h}{2} = \frac{8+15-17}{2}$

$= \frac{6}{2} = 3$

KG

3:44:49 / 11:50:56 TCS Geometry & Mensuration
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Q5. What is the measure of the radius of the circle inscribed in a triangle whose sides measure 8, 15 and 17 units?

$b = 8 \quad p = 15$
$h = 17$

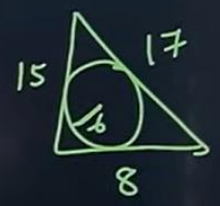
In right angled triangle, $h^2 = b^2 + p^2$

$(17)^2 = (15)^2 + 8^2$

$289 = 225 + 64$

$\boxed{289 = 289}$

$r = \dfrac{bp}{b+p+h}$



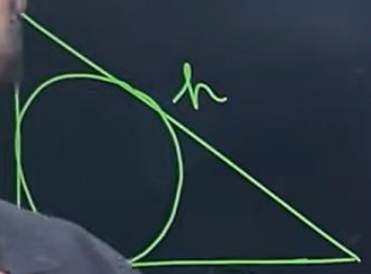
$= \dfrac{8 \times 15}{8+15+17} = \dfrac{8 \times 15}{40}$

$= 3$

$r = \dfrac{b+p-h}{2} = \dfrac{8+15-17}{2}$

$= \dfrac{6}{2} = 3$

What is the measure of the radius of the circle inscribed in a whose sides measure 8, 15 and 17 units?



$$r = \text{Inradius} = \frac{\text{area}}{\text{Semi perimeter}} = \frac{\frac{1}{2} \times b \times p}{\frac{b+p+h}{2}}$$

$$= \frac{b+p-h}{2}$$

$$r = \frac{bp}{b+p+h}$$

$$= \frac{bp}{\text{Perimeter}}$$

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